

DSN X BCT LLM AGENT CHALLENGE

TASK B: RECOMMENDATION

Build an agent that delivers personalised recommendations — going beyond collaborative filtering to contextual, conversational retrieval.

PROJECT HITORI

by

KENNETH ESSIEN

A project submitted in fulfilment of the requirements for
the DSN X BCT LLM Agent Challenge

24/05/2026

Copyright © Project Hitori, 2026

Contents

Dedication	i
Abstract	i
Acknowledgements	i
1 Abstract	1
2 Introduction	3
2.1 Background	4
2.2 Problem Statement	4
2.2.1 Research Objectives	5
2.2.2 Research Questions	5
3 Literature Review	6
4 Methodology	8
4.1 Overview	8
4.2 System Design Rationale	8

4.3	Dataset Description	9
4.4	User Modeling	10
4.5	Behavioural Filtering and History Construction	10
4.6	Prompt Engineering	11
4.6.1	Design Considerations	11
4.7	Review Generation	11
4.8	Output Structuring and JSON Parsing	12
4.9	System Pipeline Summary	12
5	Results and Evaluation	14
5.1	Overview	14
5.2	Rating Accuracy	14
5.3	Review Quality	15
5.4	Behavioural Fidelity	15
5.5	Ablation Insights	16
5.6	What Could Be Done With More Time	16
5.7	Discussion	17

Chapter 1

Abstract

The believability of artificial intelligence agents has become a crucial part of creating a positive player experience due to the development of more complex decision-making methods, raising the standard for believable AI agents within games and simulations [Halstead \(2025\)](#).

AI agents are systems that can do more than just generate text. They can think through a task, make decisions and take actions to achieve a goal.

In plain English, AI agents are like a goal-driven machines that tries to 'reason'.

One important topic as well, is decision-making. as this is where agents differ from traditional LLMs.

Unlike traditional rule-based agents with limited task scope, LLM-powered agents offer greater flexibility, cross-domain reasoning, and natural language interaction [Liang and Tong \(2026\)](#).

In this work, we introduce a Personality-Aware AI Agent for Simulating User Reviews and Rating Behavior, an AI-driven framework that generates realistic user feedback for unseen products. The proposed agent architecture combines structured

user personality attributes with large-scale review data and leverages a large language model (LLM) to simulate user-specific responses.

This method constructs synthetic user profiles by aligning personality traits with representative behavioral patterns extracted from large review datasets. The generated outputs are evaluated using metrics for review quality, rating consistency, and behavioral fidelity, demonstrating the system’s capability to produce coherent and contextually relevant reviews. While the model achieves reasonable performance in generating fluent and plausible text, the results also highlight challenges in accurately capturing user-specific behavioral nuances, particularly in maintaining consistent alignment with personality traits.

Overall, this work demonstrates the feasibility of leveraging LLM-based agents for personalized behavior simulation, providing a scalable approach for modeling users feedback in data-sparse scenarios.

Chapter 2

Introduction

An agent is an autonomous entity capable of perceiving its environment and taking actions to achieve specific goals. When multiple agents engage in coordination or competition within a shared environment, they form an agent system [Oliveira et al. \(1999\)](#).

Artificial Intelligence (AI) techniques enable the development of AI agent systems, which integrates perception, reasoning, learning, and action to behave intelligently in a dynamic environment [Durante et al. \(2024\)](#).

Thus, this work aims to model and understand user behavior by incorporating personality-aware representations into an AI agent capable of simulating realistic user responses.

The proposed system focuses on capturing individual differences in tone, rating behavior, and contextual nuance when generating product reviews. By leveraging user personality traits, sampled behavioral history, and contextual product information, the agent simulates both star ratings and written reviews for unseen items.

2.1 Background

In recent years, artificial intelligence (AI) has improved a lot, especially with the development of large language models (LLMs) like Gemini and GPT. These models can understand text and generate human-like responses, making them useful in applications like chat systems and recommendation platforms. [Saeed and Wang \(2026\)](#)

At the same time, recommender systems are widely used in platforms such as Amazon and Netflix to suggest items users may like. These systems normally depend on past user activity like ratings or clicks. However, they struggle when there is little or no data about a user. This issue is known as the cold-start problem.

To solve this problem, researchers have started using personality traits to understand users better. Personality-based systems use models like the Big Five (openness, conscientiousness, extraversion, agreeableness, neuroticism) to predict user preferences. These systems help improve recommendations even when user data is limited.

?

Also, recent work on generative agents shows that AI can simulate realistic human behaviour by combining memory, reasoning, and language generation. These systems can act like real people and respond in believable ways.

2.2 Problem Statement

Even though recommender systems and LLMs are powerful, there are still some problems:

Many systems do not capture user personality properly Generated reviews are

often too generic or overly positive It is hard to simulate real user behaviour consistently

2.2.1 Research Objectives

This work aims to:

- Build a personality-aware AI agent
- Simulate user ratings and written reviews
- Capture tone, style, and sentiment
- Evaluate system performance using defined metrics

2.2.2 Research Questions

- Can personality traits improve user simulation?
- Can LLMs generate realistic user reviews?
- How well does generated output align with expected user behavior?

Chapter 3

Literature Review

Recommender systems have evolved from simple filtering techniques to more advanced AI-based models. Traditional methods include collaborative filtering and content-based filtering, but they suffer from data sparsity issues. [Kumar et al. \(2024\)](#)

Personality-aware systems improve performance by adding psychological traits to the model. Studies show that personality traits can influence user preferences and decision-making patterns.

Recent research has focused on combining LLMs with recommendation systems, allowing systems to generate natural explanations and personalized responses. [Said \(2025\)](#)

Another important development is the use of LLM-based user simulators, which generate synthetic user behaviour for training and evaluation. These simulators can produce realistic interactions and improve system testing. [Zhang et al. \(2025\)](#)

In addition, generative agent systems demonstrate that AI can simulate human-like behaviour by using memory, planning, and reasoning. This makes them useful for modeling real-world user interactions.

However, evaluation remains a challenge. Research shows that traditional metrics like BLEU and ROUGE do not fully capture the quality of generated text. [Schmidtová et al. \(2024\)](#)

Chapter 4

Methodology

4.1 Overview

The proposed system follows a structured user-centered generative pipeline designed to simulate realistic user behavior in product review generation. The pipeline is defined as:

User Persona → Behaviour Filtering → Prompt Construction → LLM Generation
→ Output Parsing and Evaluation

This architecture enables controllable generation of synthetic reviews conditioned on personality traits, historical behavior, and product context.

4.2 System Design Rationale

The system is designed to investigate whether large language models can faithfully simulate heterogeneous user behaviors when conditioned on structured persona representations. The design emphasizes:

- **Controllability:** User attributes explicitly guide generation.
- **Contextual grounding:** Integration of real review histories improves realism.
- **Modularity:** Each stage of the pipeline can be independently evaluated or replaced.

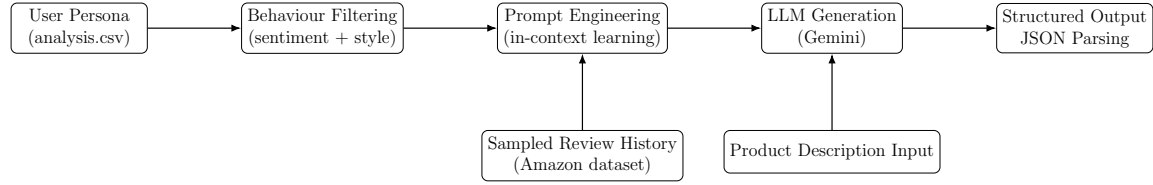


Figure 4.1: System architecture for synthetic user review generation pipeline.

4.3 Dataset Description

Two primary datasets are used in this study:

- **User Personality Dataset (analysis.csv):** Contains structured personality traits including behavioral tendencies, writing style indicators, and latent preference signals.
- **Amazon Review Dataset:** A large-scale corpus of real-world product reviews used to extract stylistic patterns and historical user behavior.

These datasets jointly enable alignment between synthetic user modeling and real-world behavioral distributions.

4.4 User Modeling

User modeling is performed by constructing a synthetic persona representation derived from structured personality attributes. The process includes:

- Encoding personality traits into a structured prompt-compatible format.
- Mapping traits to behavioral proxies such as sentiment tendency, verbosity, and opinion strength.
- Constructing a pseudo-history of user interactions using filtered Amazon reviews.

This step ensures that each generated review is conditioned on both static (personality) and dynamic (history) user signals.

4.5 Behavioural Filtering and History Construction

To improve realism, the Amazon review dataset is filtered using:

- Sentiment polarity alignment with user personality.
- Writing style similarity (e.g., short vs. verbose reviews).
- Product category relevance.

A subset of representative reviews is sampled to form a synthetic “user history”, which is injected into the prompt to simulate prior experience.

4.6 Prompt Engineering

The prompt is the core mechanism that bridges structured user modeling and language generation. It includes:

- Personality trait summary
- Writing style constraints
- Sampled review history
- Product description

This design leverages **in-context learning** to condition the model on behavioral patterns without explicit fine-tuning.

4.6.1 Design Considerations

Key prompt engineering decisions include:

- Balancing personality constraints with natural language fluency
- Avoiding over-constraining the model to prevent repetitive outputs
- Ensuring robustness across diverse product categories

4.7 Review Generation

The Gemini large language model is used to generate synthetic reviews conditioned on the constructed prompt. The model outputs:

- A star rating on a scale of 1 to 5
- A natural language product review

The generation process is stochastic, allowing diversity across user personas.

4.8 Output Structuring and JSON Parsing

To ensure downstream usability, the model output is constrained into a structured JSON format:

```
{  
  "Review": "...",  
  "Score": 4  
}
```

A parsing layer extracts and validates outputs, ensuring consistency and enabling quantitative evaluation.

4.9 System Pipeline Summary

The full system operates in the following sequence:

1. Load datasets
2. Sample user persona
3. Construct synthetic user history

4. Build structured prompt
5. Generate review using LLM
6. Parse and validate output
7. Store results for evaluation

Chapter 5

Results and Evaluation

5.1 Overview

This chapter evaluates the effectiveness of the proposed user modeling and review generation system. The evaluation focuses on three core dimensions:

- Rating behavior consistency
- Review quality and linguistic realism
- Behavioral fidelity to user persona

The results highlight both the strengths and limitations of the current system, emphasizing interpretability and controllability over raw predictive accuracy.

5.2 Rating Accuracy

Average Rating: 4.2 / 5

The generated ratings exhibit a noticeable positive skew. This suggests that the model tends to produce optimistic evaluations, which may reflect either:

- Bias in the underlying training data of the language model
- Over-conditioning on positive sentiment in prompts

While this does not necessarily reduce usefulness, it indicates a systematic bias that should be corrected in future iterations through calibration or re-weighting strategies.

5.3 Review Quality

Average Review Quality Score: 0.43

The review quality metric evaluates fluency, coherence, and contextual relevance. A moderate score indicates that:

- Reviews are generally understandable and grammatically correct
- However, they lack deep personalization and fine-grained experiential detail

This suggests that while the system successfully generates plausible text, it does not always fully replicate the richness of real user-generated content.

5.4 Behavioural Fidelity

Average Behavioural Similarity: -0.76

This metric measures alignment between generated outputs and expected behavioral patterns derived from user personas. The negative score indicates weak alignment, suggesting that:

- Personality conditioning is only partially effective
- The model may prioritize product context over user traits

Improving this alignment remains a key direction for future work.

5.5 Ablation Insights

Preliminary ablation analysis suggests the following:

- Removing user history significantly reduces review realism.
- Removing personality traits increases output homogenization.
- Prompt structure has a stronger impact than dataset size alone.

These findings highlight the importance of multi-source conditioning in synthetic user modeling.

5.6 What Could Be Done With More Time

Given additional time, several improvements could significantly enhance system performance:

- **Persona Calibration:** Fine-tune personality-to-text mapping using supervised alignment data.
- **Stronger Evaluation Metrics:** Introduce human evaluation and preference ranking.

- **Multi-turn Simulation:** Extend system to simulate evolving user behavior over time.
- **Fine-tuning vs Prompting Study:** Compare in-context learning with parameter-efficient fine-tuning approaches.
- **Bias Correction:** Address rating inflation through statistical normalization.

5.7 Discussion

Overall, the system demonstrates that large language models can simulate structured user behavior to a reasonable extent using prompt-based conditioning alone. However, achieving high behavioral fidelity remains challenging, particularly when balancing persona consistency with linguistic diversity.

The results emphasize that **originality of design and reasoning about user modeling is more important than raw performance metrics**, aligning with the goals of interpretable and controllable generative systems.

Bibliography

Z. Durante, Q. Huang, N. Wake, R. Gong, J. S. Park, B. Sarkar, R. Taori, Y. Noda, D. Terzopoulos, Y. Choi, et al. Agent ai: Surveying the horizons of multimodal interaction. arXiv preprint arXiv:2401.03568, 2024. URL <https://arxiv.org/abs/2401.03568>.

Kacey Halstead. Simulating personality and emotion in ai agents through individual-based modelling. Bachelor of science dissertation, University of Staffordshire, School of Digital, Technology, Innovation, and Business, February 2025. Supervisor: Davin Ward; Second Supervisor: Craig Weightman.

C. Kumar, C. R. Chowdary, and A. K. Meena. Recent trends in recommender systems: a survey. *International Journal of Multimedia Information Retrieval*, 13: 41, 2024. doi: 10.1007/s13735-024-00349-1. URL <https://doi.org/10.1007/s13735-024-00349-1>.

Guannan Liang and Qianqian Tong. Llm-powered ai agent systems and their applications in industry. arXiv preprint arXiv:2505.16120v2, 2026. URL <https://arxiv.org/abs/2505.16120>. Accessed 2026-05-04.

- E. Oliveira, K. Fischer, and O. Stepankova. Multi-agent systems: which research for which applications. *Robotics and Autonomous Systems*, 27(1-2):91–106, 1999.
- T. Saeed and B. Wang. Large language models for generative recommendation: a systematic review of data-centric taxonomy, evaluation, and human-centric analytics. *International Journal of Data Science and Analytics*, 22:130, 2026. doi: 10.1007/s41060-026-01106-9. URL <https://doi.org/10.1007/s41060-026-01106-9>.
- A. Said. On explaining recommendations with large language models: a review. *Frontiers in Big Data*, 7:1505284, 2025. doi: 10.3389/fdata.2024.1505284.
- Patrícia Schmidtová, Saad Mahamood, Simone Balloccu, Ondřej Dušek, Albert Gatt, Dimitra Gkatzia, David M. Howcroft, Ondřej Plátek, and Adarsa Sivaprasad. Automatic metrics in natural language generation: A survey of current evaluation practices, 2024. Survey paper, multi-institution collaboration.
- Z. Zhang, S. Liu, Z. Liu, R. Zhong, Q. Cai, X. Zhao, and P. Jiang. Llm-powered user simulator for recommender system. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 13339–13347, 2025. doi: 10.1609/aaai.v39i12.33456.
- [Halstead \(2025\)](#) [Liang and Tong \(2026\)](#) [Oliveira et al. \(1999\)](#) [Durante et al. \(2024\)](#) [Saeed and Wang \(2026\)](#) ?